

DESIGN AND DEVELOPMENT OF “YOUR DESK”: A FRONT-OFFICE OPERATIONS AND MANAGEMENT SYSTEM FOR HOSPITALITY MANAGEMENT

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ABSTRACT

Traditional front desk processes in academic settings often rely on manual methods or limited trial-based platforms, which restrict students' exposure to real-world hotel management systems. The existing systems related to hotel room reservation management are only limited to room booking and user management, lacking integrated features for operational coordination, analytics, and intelligent support. To address these issues, this study developed a web-based front-office operations and management system called "Your Desk" that integrates reservation management, inventory tracking, maintenance scheduling, calendar API for reservation synchronization, a Local AI Hotel Assistant using Ollama, and basic data analytics with seamless report data retrieval. The system aims to provide a faster, more secure, and more reliable way to manage front office tasks and streamline reservation processes during HM front desk simulation and hotel operational activities. The study used a descriptive-developmental research design guided by Agile Development methods. The system was built using the Python Django framework, MySQL database, HTML, CSS, JavaScript, and Bootstrap, while data gathering involved informal interviews and surveys for system requirements and user testing feedback. The system successfully achieved its objectives by centralizing front-office operations into a single platform that improved reservation accuracy, reduced scheduling conflicts, and enhanced inventory and maintenance monitoring. The integration of the local AI assistant supported operational efficiency, while automated reports and analytics improved decision-making and performance evaluation. In conclusion, "Your Desk" provided an effective and practical solution that bridges the gap between academic training and real-world hospitality operations. It enhanced both administrative efficiency and student learning experience, offering a reliable and scalable system for future implementation in academic institutions and hotel environments.

Keywords: front office management system, hospitality management, AI hotel assistant, analytics, technology, reservation synchronization



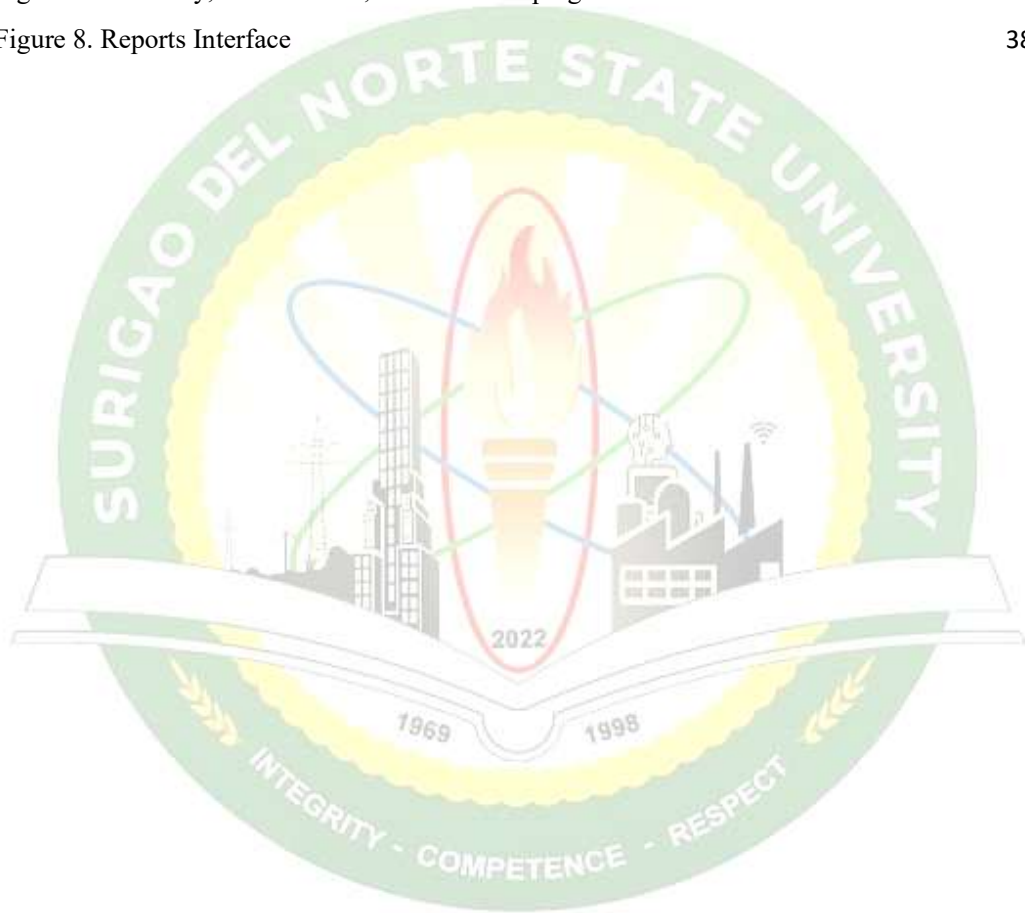
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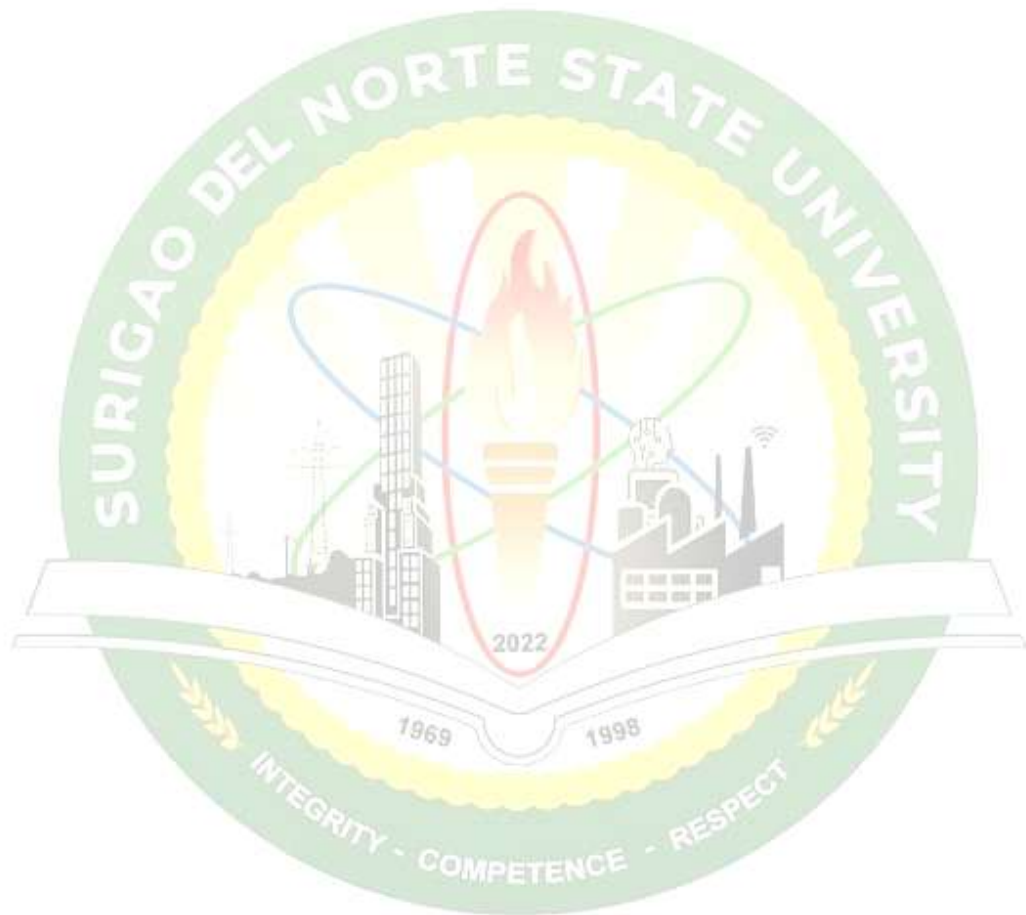
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Chapter 1

INTRODUCTION

1.1. Background of the Study

The development of “Your Desk”: A Front-Office Operations and Management System for Hospitality Management focuses on creating a web-based system that enhances front-office procedures and training within the Hospitality Management (HM) laboratory of Surigao del Norte State University. The system integrates essential hotel front-desk functions such as room reservations, guest record management, room status monitoring, inventory tracking, billing simulations, maintenance scheduling, housekeeping coordination, and automated report generation into a centralized digital platform. It is designed to simulate real-world hotel operations while replacing manual and paper-based reservation processes currently practiced in laboratory activities. The primary purpose of the study is to improve operational efficiency, ensure data accuracy, and provide a structured training environment aligned with industry standards. The system also aims to strengthen students’ digital competencies and decision-making skills in hospitality operations. For instructors and laboratory administrators, it offers improved monitoring, organized workflow management, and reliable performance evaluation tools. The importance of this study lies in its contribution to academic digital transformation and instructional innovation. By bridging theoretical knowledge and practical application, the system supports the university’s commitment to producing competent and industry-ready hospitality professionals

Several studies emphasize the significance of digital systems in improving hotel management efficiency and educational training. Studies show that hotel establishments experience challenges such as manual recordkeeping, delayed transactions, and communication gaps between staff because of not having a centralized platform for room reservation management. Through technology adoption by means of creating front-office operations and management systems, hotels improved service delivery and customer satisfaction. [1] Web-based hotel management systems reduce human errors and improve reservation accuracy through centralized databases and automated workflows. Incorporating user requirements gathered from real front-desk tasks will make the system intuitive for every stakeholders. The system must be adaptable to various hospitality workflows by incorporating an extensible database

structure and a configurable administrative module. Through an extensible database, administrators can modify or update room classifications, service categories, pricing structures, and other operational data without altering the core system architecture. This flexibility ensures that the system can accommodate differences in room types, business models, and organizational hierarchies across hospitality establishments.[2] Simulations are effective when they are intuitive and responsive. The system must include clear dashboard, visual cues, and responsive features that allow staff to perform tasks efficiently.[3]

The system is anchored on the study conducted by Alam, Rahman, and Rivinon on the development and implementation of a Python-based hotel management system, which focused on room reservation, payment, and administration. While their study provides a solid foundation for automating core hotel processes, it did not fully address advanced analytics, integration with third-party platforms, inventory and housekeeping tracking, room maintenance scheduling or the potential use of artificial intelligence to enhance decision-making and operational efficiency.[4] In addition, a significant gap remains at the SNSU institution, as there is currently no specialized front-office management system for Hospitality Management laboratories. Students rely heavily on textbooks and temporary trial versions of reservation software, which fail to provide the comprehensive, hands-on experience needed to master front-desk procedures. This gap results in inefficiencies, data inconsistencies, limited student engagement, and reduced simulation accuracy. There is a need for a customized system that aligns with curricular requirements and institutional workflows while remaining cost-effective and sustainable.[5] Data security is essential for maintaining trust and operational integrity in hospitality management systems. Protecting sensitive information safeguards the hotel's reputation and prevents potential operational disruptions caused by breaches or data loss.[6] The system incorporates features that mitigate common risks, such as improper access controls, weak authentication, or accidental data exposure through user authentication, role-based access, and secure data storage.

In conclusion, the development of "Your Desk" responds to the growing demand for a modernized and integrated front-office operations system in hospitality education. The study justifies the implementation of a centralized web-based platform

to replace inefficient manual reservation practices and disconnected administrative procedures. Through structured system development and user-centered design, the project aims to enhance training realism and operational competence among students. The system delivers functional modules for reservation management, guest profiling, billing simulation, housekeeping coordination, maintenance tracking, and automated reporting within a secure and accessible environment. It also incorporates organized access control to ensure proper user management among administrators, instructors, and students. By providing accurate data handling and real-time operational monitoring, the platform strengthens laboratory management and instructional effectiveness. The system contributes to institutional innovation and digital readiness in hospitality education. The implementation of “Your Desk” therefore represents a strategic advancement in aligning academic training with evolving hospitality industry standards.

1.2. Statement of the Problem

The Hospitality Management Laboratory of the Surigao del Norte State University currently relies on manual and fragmented tools for front-office training, which results in inefficient operations, inaccurate records, and limited exposure to modern hospitality management technologies. This study addresses the need to design and develop “Your Desk”, a web-based Front-Office Operations and Management System that automates reservations, integrates calendar-based scheduling with a local AI hotel assistant, and applies role-based access control for administrators, staff, housekeepers, and guests. It also seeks to implement automated reporting and simple data analytics to improve operational monitoring, support training activities, and enhance overall management efficiency in the hospitality laboratory.

1.3. Objectives of the Study

General Objective

This study aimed to design and develop a comprehensive web-based front-office operations and management system with automated room reservations, guest record management, inventory tracking, housekeeping coordination, and maintenance scheduling for the hospitality management laboratory of the university.

Specific Objectives

Specifically, this system:

1. Design and develop a web-based front-office operations and management system.
2. Implement a calendar-based reservation synchronization system integrated with a local AI hotel assistant.
3. Integrate inventory tracking and maintenance scheduling.
4. Generate automated reports and data analytics.

1.4. Significance of the Study

This study aimed to improve front-desk task training, operational efficiency, and digital learning experiences through the design and development of “Your Desk,” a web-based Front-Office Operations and Management System. Some hotels and universities continue to rely on manual procedures or fragmented digital tools that limit system integration, reduce data accuracy, and restrict exposure to industry-standard technologies. These limitations create gaps in operational simulation, decision-making practice, and technology-based learning in hospitality education. The proposed system provides a computerized alternative for secure, reliable, and responsive web-based platform. The project is beneficial for:

University: The system supports digital transformation by modernizing hospitality laboratory training and aligning academic instruction with current industry standards.

Hotel Establishment: The system can be used in hotels to manage front-office operations due to its extensible database structure and configurable modules.

Administrators: The system enhances monitoring, reporting, and workflow management through organized access control and real-time data management.

Staff: It reduces manual workload and allows staff to perform tasks faster and more accurately.

Students: The system strengthens practical skills, digital competence, and decision-making ability through realistic simulation of hotel front-office operations.

Future Researchers: It serves as a reference framework for developing secure, responsive, and intelligent web-based management systems.

1.5. Scope and Limitation

This study was conducted at Surigao del Norte State University (SNSU) from the 1st semester to the 2nd semester of the 2025-2026 academic year. This system uses Python Django for a web-based platform with cloud hosting, role-based access, a local hotel AI assistant API, calendar-based reservation synchronization, and an automated hotel operations system. The proposed system is tested at SNSU and evaluated by its target stakeholders for its accuracy, security, and efficiency, aiming to improve front-office training, operational simulation, and management efficiency in the hospitality laboratory.

Scope

The core features of the front-office operations and management system includes:

- 1. Design and develop a web-based front-office operations and management system.**

- 1.1. Automate room reservation and guest record management processes.
- 1.2. Provide real-time room status monitoring and room details home page.
- 1.3. Enable role-based access control for staff, housekeepers, and guests.

- 2. Calendar-Based Reservation Synchronization with a Local AI Hotel Assistant.**

- 2.1. Synchronize reservations and schedules through an integrated calendar system.
- 2.2. Integrate a local AI hotel assistant for task guidance and operational support.
- 2.3. Provide automated reminders and reservation conflict detection.

- 3. Inventory Tracking and Maintenance Scheduling.**

- 3.1. Monitor inventory usage and availability within the HM laboratory and front desk office.
- 3.2. Schedule and track maintenance activities for rooms and laboratory facilities.

- 4. Automated Reports and Data Analytics**

4.1. Generate automated operational reports such as reservation logs, inventory reports, and maintenance records.

4.2. Provide basic data analytics and downloadable reports in csv file to support monitoring and decision-making.

Limitations

However, there are certain limitations to the front-office operations and management system, including:

1. The requirement of a stable internet connection since the system operates as a cloud-based web application and cannot be accessed in offline environments
2. The system implementation covers only within the university's HM laboratory based on the concepts presented in front-desk operations textbook, which may not fully capture the complexity, scale, real-time demands of actual commercial hotel front-office operations.
3. The absence of real-time online payment services such as banking APIs or e-wallet platforms like Gcash, where payments are only recorded within the system to simulate transactions and maintain tracking for training and administrative purposes.
4. The limitations in data storage and processing capacity due to the allocated cloud hosting resources.
5. The locally integrated AI-powered hotel assistant, which may result in slower response times and does not support cloud-based AI services for faster or more scalable performance.

1.6 Definition of Terms

API. An application programming interface that enables communication between the system and external services (such as the use of AI hotel assistant) for data exchange.

Authentication. A process of verifying user identity before granting authorized access to system features.

Calendar Synchronization. A system feature that aligns reservation schedules with calendar API to prevent booking conflicts and overlaps.

Cloud-Based Storage. A remote data storage system allowing centralized access, management, and backup through internet-connected servers.

Data Analytics. The process of examining system data to generate summaries and operational reports.

Data Encryption. A security technique incorporated in the system to protect sensitive data (e.g. password) by converting it into encoded formats that can only be accessed by admin.

Django Framework. A Python-based web development framework used to build secure and scalable front-office management system web applications.

Front-Office Operations. This includes hotel management activities involving room reservation, housekeeping, guest recordkeeping, and room status monitoring.

Inventory Tracking. A feature incorporated in Your Desk System used by the staff for real-time resource monitoring and manage equipment handling/placement.

Maintenance Scheduling. A feature incorporated in Your Desk System used by the staff to organize and record maintenance task for rooms, facilities, and equipment.

RBAC. Role-based access control that restricts system access based on predefined user roles and permissions.

SDLC. A structured process used to plan, design, develop, test, and deploy software systems.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1. Local and Foreign Studies/Literature

2.1.1 Hotel Management and Reservation Systems

The hotel industry increasingly relies on digital systems to manage reservations, guest records, and operational activities efficiently. Hotel management systems are designed to automate routine front-office tasks such as room reservations, guest check-ins, billing, and reporting to improve operational efficiency and service quality. According to Weerasinghe et al., hotel management systems provide centralized platforms that allow staff to monitor guest information, room availability, and operational records in real time, thereby reducing manual errors and improving workflow efficiency [2]. Similarly, Apolinar et al. emphasized that proper system design using structured methodologies ensures that hotel information systems can effectively support decision-making and streamline hotel operations [1]. In modern hospitality environments, reservation systems are particularly important because they enable organizations to handle guest bookings, track room status, and maintain accurate records. Digital reservation systems also help reduce overbooking and scheduling conflicts through automated updates and database management. Studies have shown that the use of integrated reservation technologies significantly improves coordination between front-desk staff, housekeeping, and management departments [4]. Furthermore, digital transformation within the hospitality sector has encouraged hotels to adopt automated management tools to improve operational performance and guest satisfaction [7]. As a result, the development of efficient hotel management and reservation systems has become an essential component in modern hospitality operations.

Several studies have focused on the development and implementation of reservation and hotel management systems to improve operational efficiency. Alam et al. developed a Python-based hotel management system capable of managing reservations, payment records, and administrative

tasks within a unified platform, demonstrating that automated systems significantly improve data accuracy and operational efficiency [4]. Likewise, Geerthana and Kathiresh explored the development of a hotel reservation system designed to handle guest booking processes and manage room availability in a structured digital environment [10]. Another study by Nsama presented an online room reservation system developed for a guest lodge that allowed customers to reserve rooms remotely while enabling staff to monitor booking data more efficiently [14]. These systems demonstrate how digital tools can reduce manual workloads and improve the accuracy of hotel operations. Research conducted in Bali by Sanjaya et al. also highlighted that online reservation systems strengthen customer engagement and increase operational competitiveness within the hospitality industry [5]. However, many of these studies focus primarily on commercial hotel applications rather than educational environments such as hospitality laboratories. Additionally, most systems emphasize reservation functionality but provide limited support for training simulations or academic learning tools. Therefore, there remains an opportunity to develop systems that integrate operational automation with training-oriented features for hospitality education.

2.1.2 Digital Transformation and Artificial Intelligence in Hospitality Systems

Digital transformation has significantly influenced the hospitality industry by introducing advanced technologies that improve operational efficiency and enhance guest experiences. Modern hotel systems increasingly incorporate artificial intelligence, data analytics, and automation to streamline operational processes and support decision-making. According to Amirulloh et al., digital transformation in hospitality enables hotels to optimize operational efficiency while improving service quality through integrated management technologies [7]. Artificial intelligence plays an important role in analyzing operational data, predicting demand, and assisting hotel staff with automated recommendations. Predictive analytics tools can forecast guest preferences,

reservation patterns, and operational demands, allowing hotels to improve resource allocation and planning [8]. Furthermore, AI-driven inventory and reservation systems help manage room availability and pricing strategies dynamically based on real-time data [9]. These technologies allow hotel operators to make data-driven decisions that enhance both operational efficiency and customer satisfaction. As the hospitality sector continues to adopt intelligent technologies, integrating AI into hotel management platforms has become increasingly valuable for improving system capabilities. Consequently, AI-supported systems can assist staff in handling tasks more effectively and provide insights that support better operational management.

Various research studies have explored the integration of artificial intelligence and advanced analytics within hotel management systems. Kaul discussed how AI-driven inventory management systems can optimize hotel operations by analyzing booking patterns and predicting resource requirements, which helps hotels manage reservations and pricing strategies more effectively [9]. Similarly, Anubala and Philosophers emphasized that predictive analytics can improve hotel management by analyzing large volumes of operational data to identify trends and forecast customer behavior [8]. Research on digital market restructuring in the hospitality sector has also highlighted how hotels increasingly rely on digital platforms and intelligent technologies to remain competitive within evolving online environments [11]. Another study by Tong-On et al. demonstrated that data analytics and business intelligence tools can significantly improve operational performance in hotel businesses by providing managers with actionable insights [13]. These findings show that the integration of AI technologies can improve decision-making and operational planning within hotel systems. However, most existing studies focus on large-scale hotel businesses that utilize advanced cloud-based technologies. Limited research has explored how AI tools can be integrated into educational hospitality systems or laboratory-based training platforms. Therefore, developing a hotel management system with AI assistance

specifically designed for hospitality training environments may provide valuable educational and operational benefits.

2.1.3 System Development Practices and Software Engineering Approaches

The development of reliable software systems requires the use of structured software engineering methodologies and best practices. Software development lifecycle (SDLC) models provide a systematic approach to planning, designing, implementing, testing, and maintaining software systems. According to Leong et al., combining traditional SDLC models with agile methodologies can improve system adaptability and sustainability by balancing structured planning with iterative development processes [12]. These hybrid development approaches allow developers to respond effectively to evolving system requirements while maintaining high software quality standards. In addition, extreme programming methods have been used in the development of reservation systems to improve development efficiency and collaboration among development teams [15]. System development also involves implementing secure and scalable architectures to ensure that applications can operate reliably in real-world environments. Security considerations are particularly important when developing web-based systems that manage sensitive user information and operational data. Rayhan et al. highlighted that implementing cybersecurity best practices in Python web applications is essential for protecting user data and preventing potential vulnerabilities [6]. Therefore, applying established software engineering methodologies and security practices is critical in the development of hotel management systems.

Several research studies have investigated the design and development of hotel reservation and management systems using various software development methodologies. Sudarsono et al. applied the extreme programming approach to develop a hotel reservation system that improved development speed while maintaining system functionality and reliability [15]. Similarly, Apolinar et al. utilized the design thinking methodology in

the development of a hotel information system, demonstrating that user-centered design approaches can improve system usability and functionality [1]. Studies have also emphasized the importance of implementing proper access control mechanisms to ensure that only authorized users can access specific system features. Barkley et al. introduced the concept of role-based access control, which allows system administrators to assign different access permissions to users based on their roles within an organization [16]. Modern access control systems may also integrate advanced technologies such as blockchain to enhance security and accountability within digital platforms [17]. These approaches demonstrate that effective system development requires both structured methodologies and robust security frameworks. However, many existing studies focus primarily on enterprise-level systems rather than educational or training-focused platforms. Consequently, there is a need to explore system development strategies that address both operational automation and educational simulation within hospitality laboratories.

2.1.4 Responsive Web Systems and User Interface Design

User interface design and system responsiveness are critical factors that influence the usability and accessibility of modern web applications. Responsive design allows systems to adapt automatically to different screen sizes and devices, ensuring that users can access system features through desktops, tablets, and mobile devices. According to Parlakkiliç, responsive web design significantly improves the usability of online systems by providing flexible layouts and improved navigation across multiple devices [18]. This design approach is particularly important in hospitality systems where staff may need to access operational data from different devices during daily activities. Responsive interfaces also improve user engagement by ensuring that system content remains readable and accessible across different platforms. Madhavan emphasized that improving mobile responsiveness enhances system performance and user experience by optimizing page layouts, loading speed, and navigation efficiency [19]. These design principles ensure that systems remain accessible and functional regardless of the device used by the user. As web

technologies continue to evolve, implementing responsive interfaces has become a standard practice in modern software development. Therefore, integrating responsive design principles into hotel management systems can improve usability and accessibility for various stakeholders.

Several studies have investigated how responsive design influences the usability and effectiveness of digital platforms. Parlakkiliç conducted research on academic websites and found that responsive design significantly improved user satisfaction and accessibility during remote learning environments [18]. Similarly, Madhavan explored strategies for enhancing mobile application responsiveness and concluded that optimized interface design improves both system efficiency and user engagement [19]. These findings highlight the importance of designing systems that support multiple device types and flexible user interactions. In hospitality environments, responsive systems allow staff and administrators to access operational information conveniently while performing daily tasks. However, many existing hotel management systems still prioritize desktop interfaces and offer limited mobile compatibility. This limitation can reduce system accessibility for users who rely on mobile devices for operational tasks. Consequently, incorporating responsive design into hotel management platforms is necessary to ensure flexible and user-friendly system interaction. Such improvements can enhance the overall effectiveness of hospitality management systems used in both commercial and educational environments.

2.2. Synthesis of the Review

Both local and foreign studies emphasize improving the accuracy, reliability, and efficiency of room reservation and front-desk operation management through modern technologies. The reviewed literature and studies highlight the growing importance of digital systems in improving hotel operations, reservation management, and service efficiency within the hospitality industry. Many researchers have explored the development of hotel management systems that automate front-office processes such as room reservations, guest records, and administrative tasks. For instance, the study conducted by Alam, Rahman, Rivin, Uddin, and Sizan focused on the

development and implementation of a Python-based hotel management system that supports room reservation, payment management, and administrative monitoring within a centralized platform [4]. Other studies also emphasized the use of digital reservation systems, artificial intelligence, and data analytics to enhance operational efficiency and decision-making in hotel environments [2], [8], [9]. Additionally, several researchers examined software development methodologies, cybersecurity practices, and responsive interface design to improve the reliability, security, and accessibility of hotel management platforms [6], [12], [18]. While these studies demonstrate significant advancements in hotel management technologies, most of them are primarily designed for commercial hotel businesses and focus on operational management rather than educational or training applications. Furthermore, many existing systems prioritize reservation and transaction processing but provide limited support for simulation-based learning environments used in hospitality education. As a result, there is still a gap in developing integrated hotel management systems that combine operational automation, training simulation, and AI-assisted support specifically designed for hospitality laboratories.

This study addresses the identified gap by proposing a Front-Office Operations and Management System designed to support both operational management and hospitality laboratory training. While the study of Alam et al. [4] demonstrated the effectiveness of a Python-based hotel management system for handling reservations, payments, and administrative tasks, the current study expands this concept by incorporating additional features tailored for educational use. The proposed system integrates modules for reservation management, guest records, housekeeping coordination, maintenance tracking, and inventory monitoring within a unified platform. In addition, the system includes role-based access control to simulate real-world hotel staff responsibilities such as administrators, front-desk staff, housekeepers, and guests. Unlike many existing studies that focus on operational business systems, this research introduces simulation tools that allow hospitality students to practice front-office operations in a controlled learning environment. The system also incorporates an AI-powered local hotel assistant to support task guidance and improve system interaction during training activities. Furthermore, responsive interface design ensures that the system can be accessed across multiple devices to enhance usability

and accessibility. Hence, this study proposes the development of a Front-Office Operations and Management System with AI-assisted support to improve operational simulation, management efficiency, and training effectiveness within the Surigao del Norte State University Hospitality Management Laboratory.



Chapter 3

THEORETICAL AND CONCEPTUAL FRAMEWORK

3.1. Theoretical Framework

Developing an intuitive web-based system for managing front-office operations and enhancing the training of hospitality management students in academic institutions is grounded in key concepts from software development, security, and online data management. This study is anchored on three major theoretical foundations: Human-Computer Interaction (HCI), Hospitality Operations Management Principles, and Cloud Computing Models. Human-Computer Interaction emphasizes the design of interactive systems that are efficient, understandable, and accessible to users with varying levels of technical experience. In the context of hospitality laboratories, HCI principles ensure that the interface allows students and instructors to easily navigate system modules such as reservations, guest records, and room monitoring. Effective interaction design also supports the learning process by enabling users to simulate front-office procedures with minimal difficulty. The system incorporates responsive web design to ensure compatibility with desktops and mobile devices, allowing users to interact with the system in a flexible environment. Through HCI-based design practices, the system aims to improve usability, reduce operational errors, and support the training needs of hospitality management students. These concepts provide a theoretical foundation for developing an intuitive and user-centered front-office operations platform.

Another theoretical foundation of this study is Hospitality Operations Management Principles, which describe how hotel front-desk operations coordinate guest services, reservations, room management, and communication among departments. In the hospitality industry, the front office acts as the central hub that manages guest interactions and operational information. The system developed in this study reflects these operational principles by integrating modules for reservations, guest information management, housekeeping coordination, maintenance tracking, and inventory monitoring. These modules simulate real-world hotel processes, allowing hospitality students to understand the workflow and responsibilities of front-office personnel. Role-based access control is also implemented to represent different operational roles such as administrators, staff, housekeepers, and guests within the

system. By modeling the operational structure of a hotel environment, the system enables users to experience realistic scenarios that mirror professional hospitality practices. The integration of these operational components helps students develop a deeper understanding of hotel management procedures and service coordination. Therefore, hospitality management principles serve as a framework for structuring the operational features of the system.

The system architecture is also supported by Cloud Computing Models, which provide scalable and centralized data management for modern web applications. Cloud-based systems allow data such as reservation records, room status updates, and operational reports to be stored and accessed securely through the internet. This framework supports real-time data synchronization and ensures that system information remains updated across multiple user roles and devices. The system utilizes cloud hosting to enable continuous access for students and instructors who participate in hospitality laboratory training. In addition, several algorithms are integrated to support automated system functions and operational decision-making. These include AI chat integration for assisting users with hotel-related tasks, data export algorithms for generating downloadable reports, and payment processing algorithms for recording financial transactions. Additional algorithms such as room availability checking, booking conflict detection, price calculation, and status ordering help maintain accurate operational records. Search and filtering algorithms further improve system usability by allowing users to quickly locate relevant information within the database. Through these cloud-based and algorithm-driven processes, the system ensures efficient management of hospitality operations and training activities.

The development of the front-office operations and management system follows the Agile Software Development Life Cycle (SDLC), which emphasizes iterative development, continuous improvement, and collaboration between developers and users. Agile development allows system features to be refined gradually through repeated cycles of design, implementation, and evaluation. During the development process, feedback from users helped improve the system interface, terminology, and functional design to better align with hospitality industry standards. The system was also evaluated using criteria based on ISO/IEC 25010 software quality standards to assess its performance, usability, reliability, security, and efficiency. User testing

allowed the developers to identify areas for improvement and refine system features to enhance its functionality and training effectiveness. The results of the testing process guided the enhancement of the web interface and the refinement of key operational modules within the system. By combining Agile development practices with standardized software quality evaluation, the system ensures that both technical performance and user experience are effectively addressed. This theoretical framework therefore supports the development of a reliable, user-centered, and educationally relevant front-office operations and management system.

3.2. Conceptual Framework

The conceptual framework of this study demonstrates how the input, process, and output (IPO) diagram works and how its components interact. The research process is structured around a conceptual model that outlines the flow of data and the development stages, as illustrated in the study's IPO Model (Figure 1).

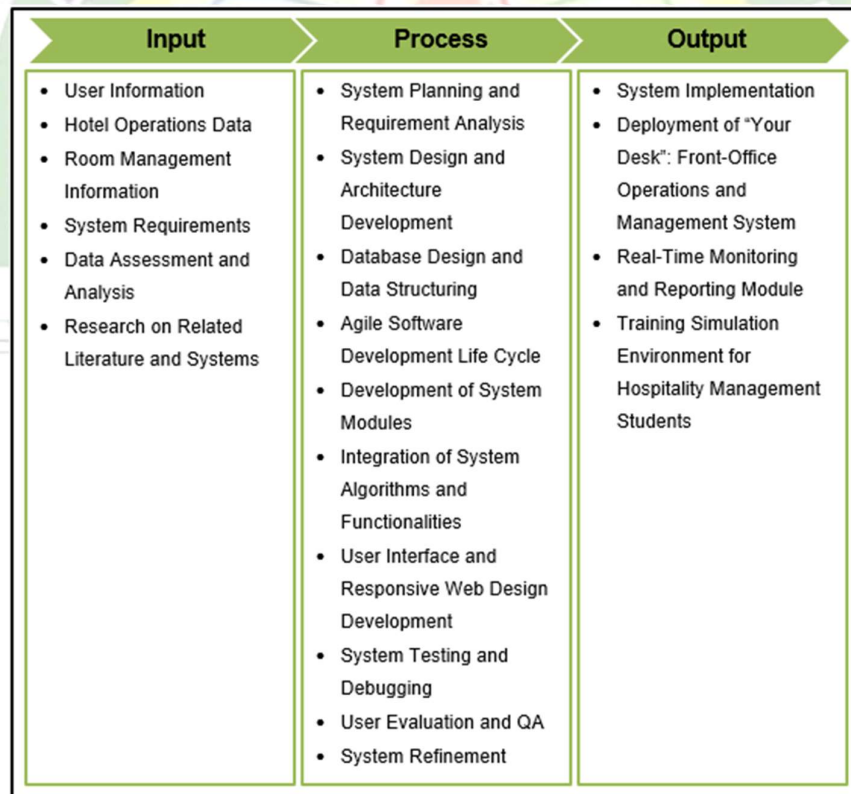


Figure 1. IPO Model of the Study

As shown in Figure 1, the system begins with gathering the necessary inputs that will guide the design and functionality of the system, which involves creating user

information, incorporating hotel operations data, managing room reservation information, analyzing system requirements, drafting a scalable system data structure, and researching on related literature and existing systems. Furthermore, the assessment and analysis of the obtained data ensure that the system aligns with the stakeholders' requirements and operational goals. The process stage involves a series of systematic development activities such as system planning and requirement analysis, system design and architecture development, database design and data structuring, and implementation using the Agile Software Development Life Cycle. It also includes the development of system modules, integration of system algorithms and functionalities, user interface and responsive web design development, system testing and debugging, user evaluation and quality assurance, and continuous system refinement. These processes ensure that the system is functional, reliable, and aligned with the operational needs of the hospitality laboratory. The output phase represents the final result of the development process, which includes the system implementation and deployment of "Your Desk": A Front-Office Operations and Management System. The system provides real-time monitoring and reporting capabilities and serves as a training simulation environment for hospitality management students to practice front-office operations in a structured and technology-supported setting. A crucial component of the framework is feedback, which is gathered through user evaluation, informal interviews and quality assurance surveys where students and instructors assess the system by answering the System Usability Scale (SUS) questionnaire, enabling continuous improvement and refinement of the system.

3.3. System Specification

This section describes the system's architecture, tools, and functional components.

3.3.1. System Architecture

The architecture follows a three-tier model consisting of the presentation layer, application logic layer, and data management layer:

- 1. Frontend (Presentation Layer).** Developed using HTML, CSS, and JavaScript with styling framework like Bootstrap to create a responsive and user-friendly interface. It provides users with a seamless experience across different devices.

2. **Backend (Application Logic Layer).** Utilized Django framework with REST API routes that manage server-side logic and communication between the interface and the database. It handles authentication, role-based access control, reservation processing, data validation, AI assistant integration, and operational analytics to ensure that system functions operate securely and efficiently.
3. **Database (Data Management Layer).** Built with MySQL that ensures secure, efficient data retrieval and structured data storage for user accounts, reservations, room details, inventory lists, housekeeping information, maintenance scheduling, and operational logs.

3.3.2 Development Tools and Technologies

The selection of modern frameworks and libraries used to build the system's infrastructure is summarized in the Development Tech Stack in Table 1:

Table 1. Development Tech Stack

Category	Tools/Technologies
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Django, Python, REST API
Database	MySQL
Media Management	Django Media Files, File Storage System
Authentication	Django Authentication System, Role-Based Access Control (RBAC)
Hosting and Deployment	PythonAnywhere
Version Control	Git & GitHub

As shown in Table 1, the combination of these development tools enables the creation of a responsive, secure web-based front-office operations and management system. Using widely adopted programming languages and frameworks ensure compatibility across different browsers and devices without requiring additional software installation. The backend relies on the Django framework and Python, supported by REST API routes to handle server-side logic, authentication, and integration with local AI features. The selected database and server technologies enable real-time data storage, retrieval, and reporting to meet the system's performance goals.

Media files are managed through Django Media Files and a dedicated file storage system to ensure proper handling of images and documents. User authentication and role-based access control (RBAC) guarantee that only authorized users can access specific system functionalities. Furthermore, the system is hosted and deployed on PythonAnywhere with version control maintained through Git and GitHub, ensuring secure deployment, collaborative development, and efficient system updates.

3.3.3 Functional Components

The system is composed of several functional components, each designed to handle specific operations to ensure efficient and secure management of hotel activities.

1. **User Authentication Module** – Ensures secure login and enforces role-based access control for different user types.
2. **Room Reservation Module** – Manages room bookings with calendar-based reservation synchronization to prevent conflicts.
3. **Payment Module** - Records payment transactions for auditing and data analytics
4. **Housekeeping Module** - Organizes and tracks room cleaning schedules and staff assignments.
5. **Inventory Module** – Monitors and manages supplies and stock levels for smooth hotel operations.
6. **Maintenance Module** – Schedules and records maintenance tasks to keep facilities in optimal condition.
7. **Management Module** - Manages user information, room details, and pricing rates for administrative control.
8. **Analytics and Reports Module** – Provides a real-time dashboard with system insights and generates reports in CSV format.

3.3.4 Software Requirements

The software environment necessary for the development and operation of the Front-Office Operations and Management System is presented in Table 2.

Table 2. Software Requirements of the Your Desk System

Software Component	Description
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Windows 10 or higher	Used for system development, deployment, and administration
Google Chrome, Mozilla Firefox, Microsoft Edge	Provides cross-browser access to the web-based system
HTML, CSS, JavaScript	Used for system logic, interface design, and interactivity
MySQL	Stores user data, room reservations, payment records, and operational information.
Visual Studio Code	Used for writing and managing system source code
XAMPP	Provides a local development environment with Apache server, PHP, and MySQL for testing the system.
Git	Version control system for source code management and collaboration.

The software requirements in the Table 2 ensure that the Front-Office Operations and Management System operates efficiently in a modern web-based environment. By using widely supported browsers such as Google Chrome and Mozilla Firefox, the system can be accessed without additional local installations. The use of common programming languages such as HTML, CSS, and JavaScript allows for seamless interactivity and consistent interface design across platforms. Additionally, the reliance on standard database management tools ensures that room reservations, payment records, and other operational data are stored securely. Tools like Visual Studio Code and Git support development and collaboration by managing source code effectively. XAMPP provides a local server environment that enables developers to test and debug the system before deployment. These software provides the necessary stability for real-time monitoring and secure record-keeping.

3.3.5. Security and Data Privacy

The system employs specific protocols to safeguard user information and maintain data integrity, as summarized in Table 3.

Table 3. Security and Data Privacy Measure

Security Aspect	Description
User Authentication	Verifies user login credentials and authenticates staff, housekeepers, and guests.
Role-Based Access Control	Restricts system features and data access based on user roles using session information.
Data Encryption	Encrypts sensitive data such as passwords and personal information using secure algorithms like SHA-256.
Input Validation	Validates all inputs to prevent invalid data, SQL injection, and other common security attacks.
Secure Database Access	Limits database connections to authorized applications and uses parameterized queries to prevent unauthorized access.
Privacy Compliance	Ensures personal data is handled in accordance with applicable privacy laws and regulations

The security and data privacy measures presented in Table 3 are implemented to protect user information and ensure the overall integrity of the system. User authentication verifies login credentials to prevent unauthorized access to the system. Role-based access control limits system features and data access according to user roles, enhancing operational security. Data encryption safeguards sensitive information such as passwords and personal details using strong algorithms like SHA-256. Input validation prevents invalid data entry and defends against common attacks, including SQL injection. Secure database access and privacy compliance maintain proper handling of information and adherence to legal data protection requirements. These combined security layers ensure that the university remains compliant with data

protection standards while providing a safe and reliable environment for staff, housekeepers, and guests.

3.3.6. Use Case Diagram

To define the functional boundaries and user roles within the systems, the interactions between students and administrators are mapped in the use case diagram shown in Figure 2.

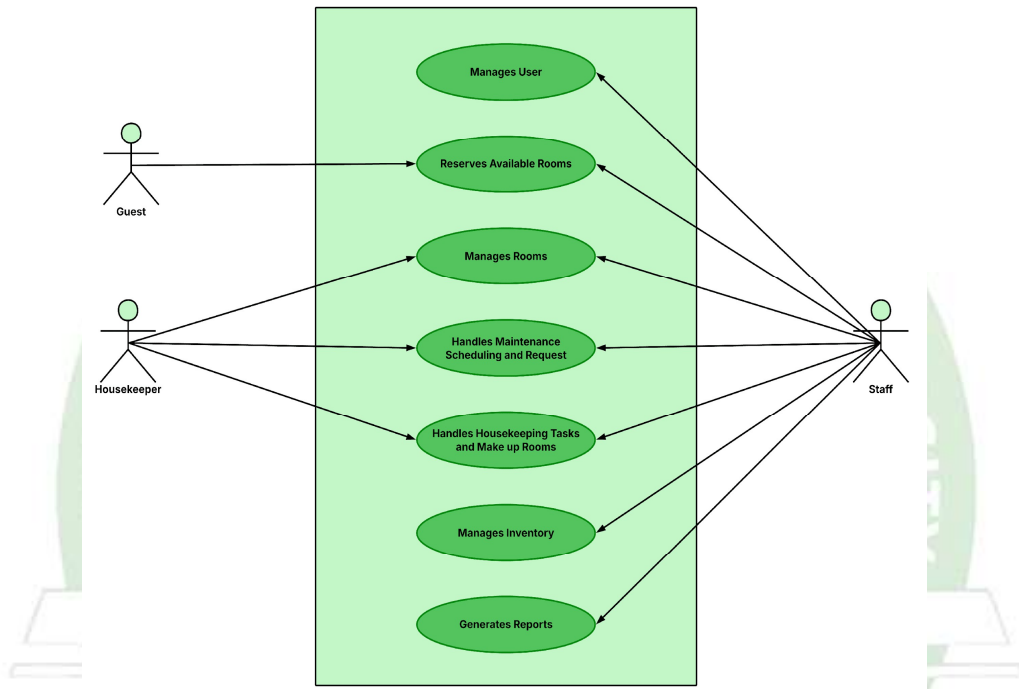


Figure 2. Use Case Diagram for Your Desk System

The diagram in Figure 2 illustrates that while guests focus on reserving rooms and browsing room details, staff manage the high-level tasks of overseeing reservations, inventory, and operational workflows. Guests interact with the system primarily through the “Reserve Available Rooms” use case, enabling them to book rooms and check availability efficiently. Housekeepers engage with functions such as managing rooms, handling maintenance requests, and performing housekeeping tasks to ensure rooms are ready for guest use. Staff have broader access, including managing users, coordinating room operations, controlling inventory, and generating performance reports. Each actor operates under a Role-Based Access Control structure, which restricts access to functions according to their responsibilities. The system boundary encompasses all core processes, indicating that every use case within the boundary is

automated by the Front-Office Operations and Management System. Collectively, these interactions streamline hotel operations, enhance guest satisfaction, and ensure that internal administrative and maintenance tasks are carried out effectively. The diagram highlights the collaborative nature of the system, supporting both customer needs and staff workflows.

3.3.7. System Design

The technical architecture and modular breakdown of the platform, including the integration of external APIs, are detailed in the low-level system design in Figure 3.

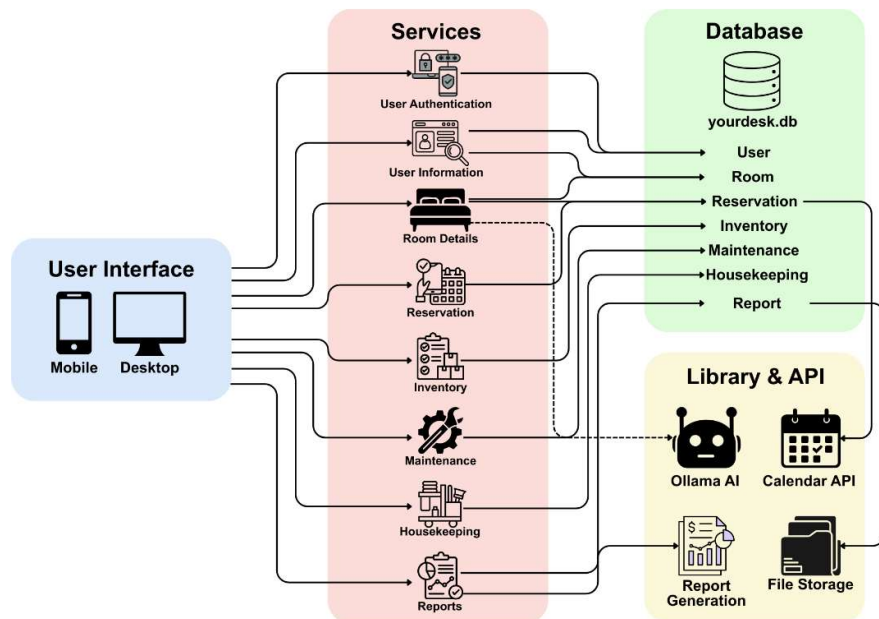
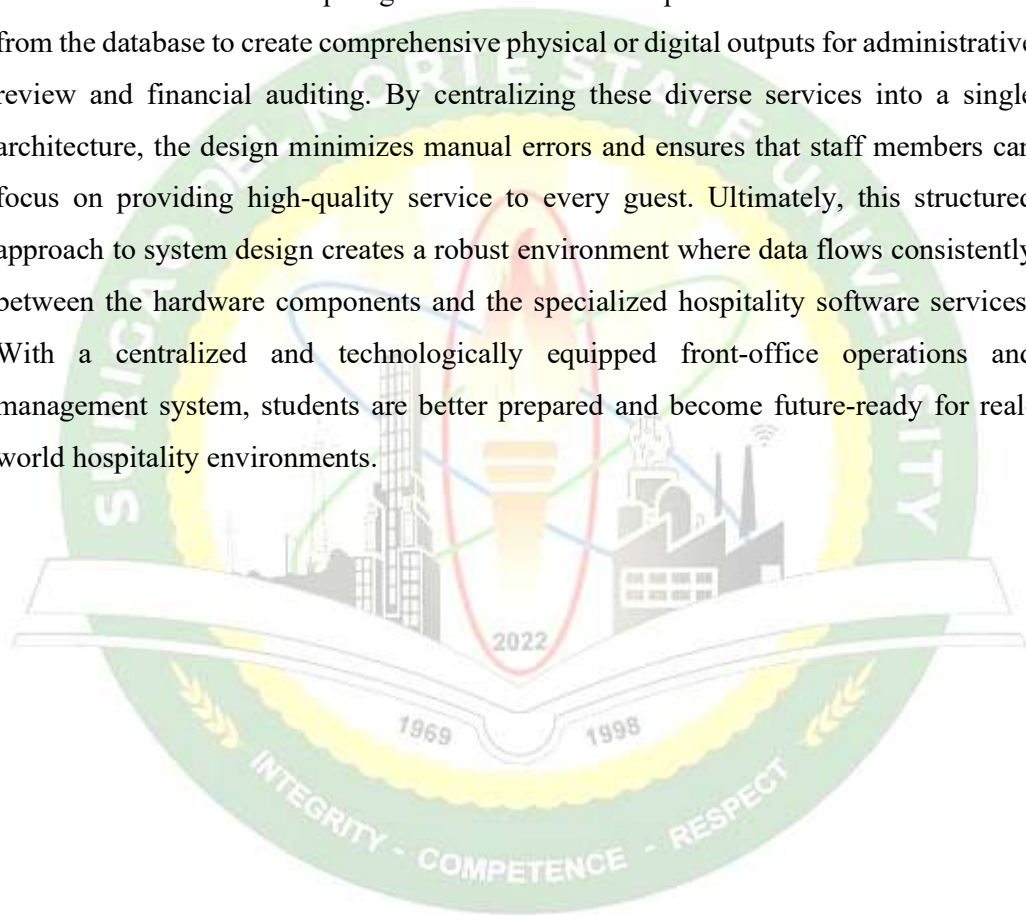


Figure 3. Low-level System Design of the Your Desk System

According to Figure 3, the service module ensures that all essential guest and administrative functions are processed accurately and efficiently within the unified digital framework. This system comprises multiple components including the user interface, services, library & API, and database which all work together to provide a seamless front office task experience and efficient reservation process. The service module is responsible for managing user registration, authentication, and role assignment to ensure that each user's access is restricted based on their privileges. Additionally, the room management, reservation and payment processing responsibilities falls to the staff to oversee pending reservation request and

housekeeping activities to reduce conflict and generate more profit. The integration of the Ollama AI through the library and API section allows the system to provide intelligent assistance and automated insights for better decision-making during peak hours. Furthermore, the Calendar API works in connection with the reservation module to synchronize all booking dates and prevent any potential scheduling conflicts or double-booked rooms. All processed data including maintenance logs and guest profiles is moved through the system bus to be stored within the shared central memory for real-time access. The report generation sub-module pulls this archived information from the database to create comprehensive physical or digital outputs for administrative review and financial auditing. By centralizing these diverse services into a single architecture, the design minimizes manual errors and ensures that staff members can focus on providing high-quality service to every guest. Ultimately, this structured approach to system design creates a robust environment where data flows consistently between the hardware components and the specialized hospitality software services. With a centralized and technologically equipped front-office operations and management system, students are better prepared and become future-ready for real-world hospitality environments.



Chapter 4

RESEARCH METHODOLOGY

4.1. Research Design

The study utilized a descriptive-developmental research design that focused on the design, development, deployment, and evaluation of a web-based front-office operations and management system for Surigao del Norte State University. The study followed the principles of the Agile Software Development Life Cycle to ensure flexibility, continuous improvement, and efficient system development through iterative testing and refinement. This research design allowed the researchers to systematically analyze existing front-office procedures and identify system requirements necessary for the development of the proposed platform. The development phase involved designing the system architecture, creating the database structure, and implementing core modules for reservation management, guest records, housekeeping coordination, and operational monitoring. The system was then deployed through the cloud hosting platform PythonAnywhere and implemented within the Hospitality Management Laboratory to simulate real-world front-desk operations in an academic training environment. To determine the effectiveness and reliability of the system, user evaluation was conducted among selected students and instructors who interacted with the platform during testing. The respondents assessed the system by answering the System Usability Scale questionnaire, while both quantitative and qualitative data were collected to evaluate the system's usability, functionality, and overall performance in addressing the identified operational issues.

4.2. Research Respondents

The target respondents of the study are currently enrolled students at Surigao del Norte State University, particularly those in the Hospitality Management and Computer Science programs, who participated in front-desk training and are expected to interact with the Front-Office Operations and Management System during the pilot and evaluation phases. Computer Science students are included to assess the technical aspects, system design, and overall software performance, while Hospitality Management students evaluate the accuracy and practicality of front-office operations. Faculty members and HM laboratory staff are also involved to provide expert feedback on operational workflows and training outcomes. They are chosen to provide direct

feedback on the system's usability, functionality, and performance under realistic classroom and simulated front-office conditions. A purposive sampling technique was employed to ensure that the selected respondents were directly relevant to the study's objectives. This method allows researchers to focus on individuals who can provide meaningful insights into the system's core workflows, accessibility, and overall efficiency. The sampling strategy also ensures representation across various year levels, academic backgrounds, and practical experience levels.

Inclusion criteria require that respondents must (1) be officially enrolled students at Surigao del Norte State University, (2) have completed or be currently participating in front-desk training, and (3) be willing to actively engage with and provide feedback on the system. Meanwhile, exclusion criteria eliminate incomplete, duplicate, or invalid entries to maintain data reliability and integrity. For the qualitative component, only respondents who have actively used the system during the pilot phase are included to capture in-depth experiences and insights. Staff and instructors are asked to evaluate both technical and operational aspects of the system to identify potential gaps or improvements. The sample size is determined to balance comprehensive feedback with feasibility of data collection and analysis. Respondents are assured of confidentiality and the voluntary nature of their participation to encourage honest and accurate responses. Finally, the combined perspectives of students, faculty, and staff provide a holistic view of the system's effectiveness, usability, and potential for broader implementation.

4.3. Research Instruments / Tools

The study utilized several research instruments to collect data and feedback during the system testing and evaluation phases. These instruments include survey questionnaires, structured interview guides, and system-generated logs. The survey questionnaire was developed based on the System Usability Scale (SUS) and other standardized usability metrics to evaluate system usability, efficiency, security, and analytics capabilities. The structured interview guide was designed to gather qualitative insights from students, instructors, and staff, focusing on their experiences with system setup, user management, and operational challenges. System-generated logs were analyzed to capture actual usage data, including check-in and check-out times, transaction accuracy, error occurrences, and system response times. These combined

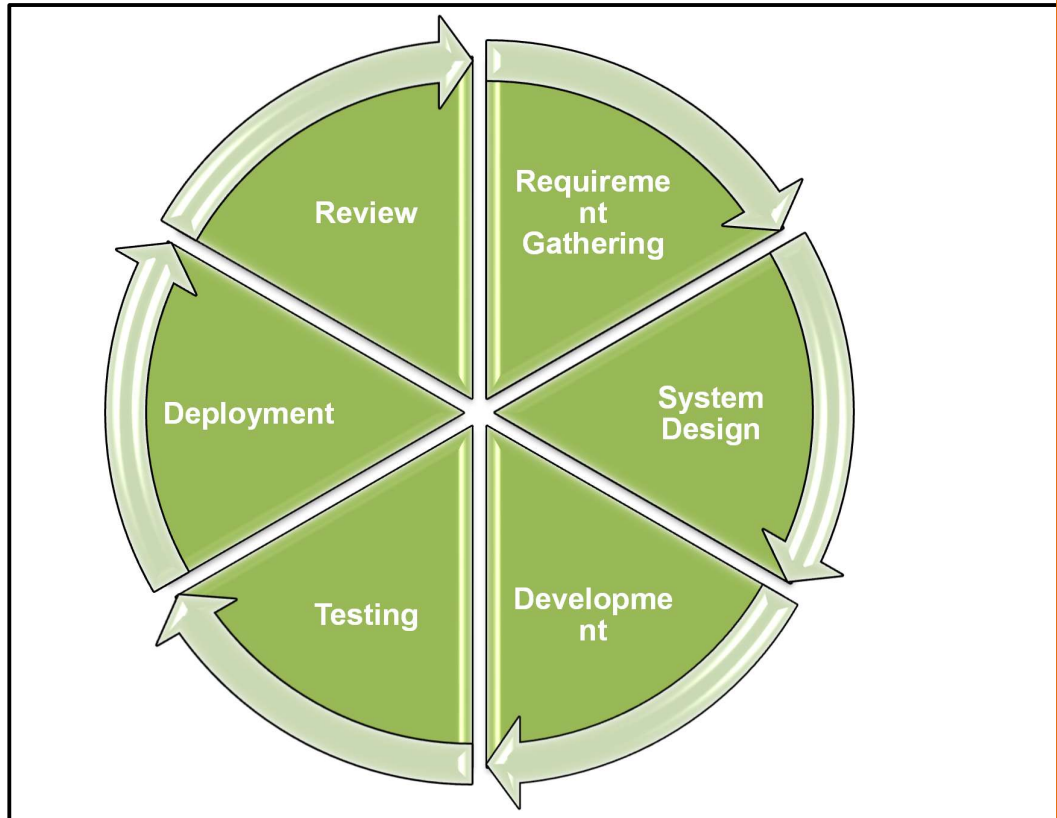
data sources provided a comprehensive understanding of user satisfaction, system functionality, and operational performance. They also helped identify areas for improvement and informed recommendations for future development and deployment.

The study also utilized technical tools essential for system implementation and analysis. Unified Modeling Language (UML) diagrams were used to visually represent system workflows, data interactions, and user scenarios to ensure clarity in design and evaluation. MySQL served as the backend database management system to securely store user information, transaction records, and operational data for analysis. The Python Django framework was employed to develop the web-based system, allowing for efficient integration of front-office modules, user interfaces, and automated functionalities. These tools facilitated accurate data collection, real-time monitoring, and robust system testing. Additionally, the system incorporated built-in analytics and reporting features to provide continuous performance feedback. Together, the combination of surveys, interviews, system logs, UML diagrams, MySQL, and Django ensured that both qualitative and quantitative aspects of system evaluation were thoroughly addressed. This multi-tool approach strengthened the reliability, validity, and overall comprehensiveness of the research findings.



4.4 Research Implementation (SDLC Phases)

The development process of Your Desk system followed an Agile Software Developmental Model to ensure flexibility, iterative improvement, and continuous stakeholders' feedback throughout the project lifecycle. The model adopted in this



study is illustrated in Figure 4.

Figure 4. Agile Software Developmental Model of Your Desk System

As shown in the Figure 4, this model allowed continuous interaction between developers and users throughout the software development life cycle. The process began with requirement gathering, where researchers collected functional and non-functional requirement from HM instructors to understand the needs of front-office operations. The system design and prototyping phase followed, UML diagrams, user interface layouts, and database schemas were created to map workflows, user interactions, and data structures aligned to the stakeholders' system requirement. During development, the system was built using the Python Django framework and MySQL database, integrating all front-office modules such as reservations, guest records, maintenance scheduling, and inventory and housekeeping tracking. The

development phase implemented these designs into a fully functional system through iterative cycles, ensuring that each version improved based on gathered input. In the testing phase, the system underwent pilot testing with target users to identify bugs, measure usability, and evaluate performance under real-world conditions. Following successful testing, the deployment phase implemented the system for broader use by deploying onto cloud hosting website and within the HM laboratory, allowing students and staff to perform actual front-office operations through local or online platforms. Finally, the review phase involved analyzing system logs, user feedback, and survey data to assess overall effectiveness, identify areas for improvement, and refine the system for optimal performance and sustainability.

4.4.1. Requirement Gathering

The requirement gathering phase involved collecting functional and non-functional data from students, HM instructors, and staff to identify the core needs of the Your Desk System. Interviews and informal discussions helped identify essential features, including automated guest check-in and check-out, real-time room availability tracking, reservation management, inventory monitoring, and housekeeping coordination. Key requirements included the need for a user-friendly interface that allows students and staff to access and update records efficiently while minimizing manual errors. Stakeholders also requested dashboards that display integrated analytics, such as peak check-in periods, occupancy rates, and operational alerts for maintenance or housekeeping tasks. Faculty and staff emphasized the importance of generating reports that match institutional standards and include verification details for auditing and monitoring purposes. Additionally, security features such as role-based access, data encryption, and activity logs were highlighted as critical to protect sensitive information. These comprehensive requirements ensured that the system was aligned with the operational workflows and management needs of Surigao del Norte State University, providing a foundation for the design and development phases.

4.4.2. System Design

The system design phase of the Your Desk system focused on translating the gathered requirements into detailed specifications and visual representations to guide development. UML diagrams, including use case, class, activity, and system architecture diagrams, were created to illustrate system workflows, user roles, and data

interactions. User interface layouts and wireframes were developed to ensure intuitive navigation and efficient access to front-office operations such as reservations, guest records, inventory management, housekeeping, and maintenance scheduling. Database schemas were designed using MySQL to define tables, relationships, and data integrity rules for secure and organized information storage. Prototyping allowed stakeholders, including HM instructors and staff, to interact with preliminary versions of the system and provide feedback on usability and functionality. The design phase emphasized alignment with stakeholder requirements, operational efficiency, and scalability for future system enhancements. Overall, the System Design phase served as a blueprint that ensured the development team could build a fully functional, user-centered, and reliable front-office operations and management system.

4.4.3. Development

The development phase of the Your Desk system involved building a fully functional Front-Office Operations and Management System using the Python Django framework and MySQL database. During this phase, all front-office modules, including reservations, guest records, inventory tracking, housekeeping coordination, and maintenance scheduling, were integrated into a single platform. The development process followed iterative cycles, allowing continuous refinement of system features based on feedback from HM instructors and student users. A comprehensive analytics suite was implemented to capture operational data, monitor system usage, and generate performance reports in real time. User interfaces were designed to be intuitive and responsive, ensuring ease of navigation for both students and staff during actual front-office operations. Security features, such as role-based access control and data validation, were embedded to protect sensitive information and maintain data integrity. Overall, this phase translated system designs and prototypes into a reliable, efficient, and user-centered platform ready for testing and deployment.

4.4.4. Testing

During the testing phase, the Your Desk system underwent comprehensive evaluation to ensure reliability, usability, and accuracy of all front-office operations. Pilot testing was conducted with selected students, instructors, and staff who interacted with the system under realistic operational scenarios. Functional testing verified that modules such as reservations, guest records, maintenance scheduling, and inventory and housekeeping tracking performed as intended without errors. Usability testing

assessed the intuitiveness of the user interface, navigation, and accessibility to ensure a seamless experience for end-users. System-generated logs and built-in analytics were utilized to monitor real-time performance metrics, including transaction accuracy, response times, and error occurrences. Feedback gathered from surveys and structured interviews was incorporated to identify potential improvements and optimize workflows. This iterative testing process, combined with the comprehensive analytics suite, allowed the developers to refine the system for stability, efficiency, and alignment with stakeholder requirements before full deployment.

4.4.5. Deployment

The deployment phase involved making the Your Desk system fully operational and accessible to its intended users. The system was deployed both on a cloud hosting website and locally within the SNSU HM Laboratory, ensuring flexibility for online and offline use. Cloud deployment allowed remote access for students, instructors, and staff, while the local installation facilitated hands-on training in the laboratory environment. During deployment, all front-office modules—including reservations, guest records, maintenance scheduling, inventory tracking, and housekeeping management—were fully integrated and tested for reliability. A comprehensive analytics suite was also implemented to monitor system usage, track performance metrics, and generate operational reports in real time. Training sessions and user orientation were conducted to familiarize stakeholders with system functionalities and features. Overall, this phase ensured that the system was operational, accessible, and capable of supporting real-world front-office workflows while providing continuous data for evaluation and improvement.

4.4.6. Review

In the review phase, researchers evaluated the overall performance, usability, and effectiveness of the Your Desk system using multiple assessment tools. A System Usability Scale (SUS) questionnaire was administered to students, instructors, and staff to measure satisfaction, ease of use, and overall user experience. In addition, a comprehensive analytics suite embedded within the system was used to capture real-time data on user interactions, transaction accuracy, error occurrences, and system response times. The collected survey and analytics data were analyzed to identify patterns, operational gaps, and areas requiring improvement. Feedback from this phase was used to prioritize refinements in system features, interface design, and workflow

automation. The iterative review ensured that both qualitative and quantitative insights informed system optimization. Ultimately, this phase provided a foundation for continuous enhancement, ensuring that the system could reliably support front-office operations while meeting stakeholder expectations.

4.5 Research Deployment (System Testing, User Acceptance)

The deployment phase involved implementing the Your Desk system for actual use while ensuring its reliability and effectiveness through rigorous testing. Initially, pilot testing was conducted with a selected group of students, instructors, and staff to evaluate system functionality, identify bugs, and verify that all front-office modules operated as intended. Following successful pilot testing, the deployment setup was completed by hosting the system on a cloud platform and configuring it for access both locally within the HM laboratory and remotely for online users. The User Acceptance Testing (UAT) process was then conducted, allowing end-users to perform typical front-office operations such as reservations, guest record management, inventory tracking, and maintenance scheduling. Feedback gathered during UAT focused on usability, workflow efficiency, and overall system satisfaction, ensuring that the system met stakeholder requirements. Any issues identified during this phase were documented and addressed through iterative improvements to optimize performance and reliability. Ultimately, this deployment strategy confirmed that the system was fully operational, user-friendly, and ready for broader implementation within Surigao del Norte State University's front-office training environment.

Chapter 5

RESULTS AND DISCUSSIONS

This chapter presents the results and discussion of the developed Your Desk Front-Office Operations and Management System for Surigao del Norte State University. The presentation is organized based on the four specific objectives and their corresponding scopes.

5.1. Front-office operations and management system

The system's administrative control center is accessed via a secure portal that provides various front-office features such as room reservation details, and user role assignment as shown in the Home and User Management Interface in Figure 5.

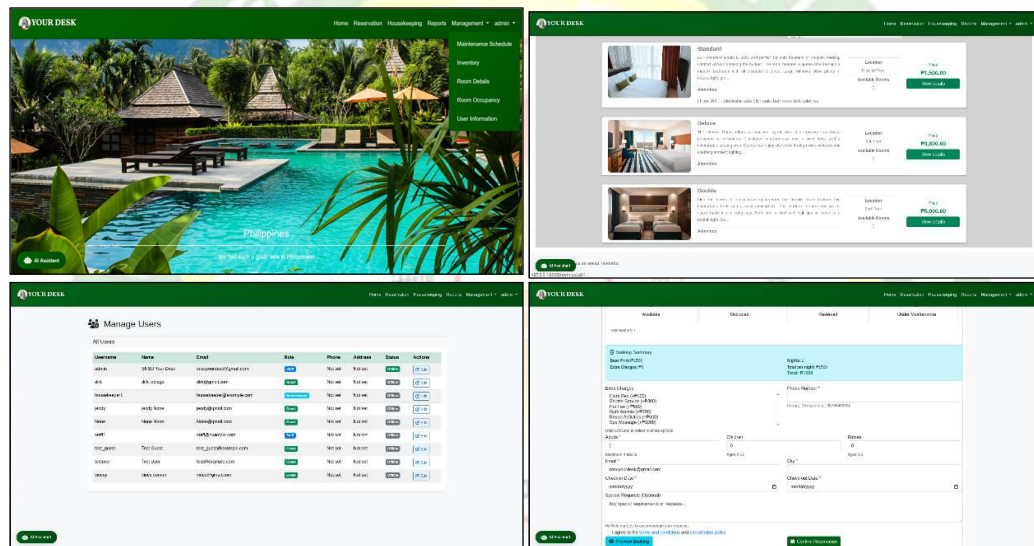


Figure 5. Home and User Management Interface

The development of this feature successfully meets the project scope for centralized front-office operation management and secure data handling. As illustrated in Figure 5, the interface enables staff to automate room reservations and efficiently manage guest records. Users can view real-time room availability and details, ensuring accurate monitoring of hotel room occupancy. Role-based access control restricts features according to user roles, allowing staff, housekeepers, and guests to access only relevant functions. This implementation aligns with the first objectives by streamlining administrative tasks, improving operational efficiency, and maintaining data security. Compared to related systems, the Your Desk interface provides a more integrated and user-friendly approach by combining reservation management with user administration

in a single platform. Web-based access eliminates the need for manual reservation process which allows hotel establishment administrators to update room details from any location and keep records for auditing process. The automated workflows reduce manual errors and enhance overall system reliability. Furthermore, the system ensures that only assigned staff have the authority to modify operational logs, maintaining the integrity of the hotel establishment and university training records.

5.2. Calendar-Based Reservation Synchronization with a Local AI Hotel Assistant

To synchronize reservation schedules on calendar visualization, integrate a local AI hotel assistant for operational support, and provide automated reminders for minimized reservation conflicts, as illustrated in Figure 6.

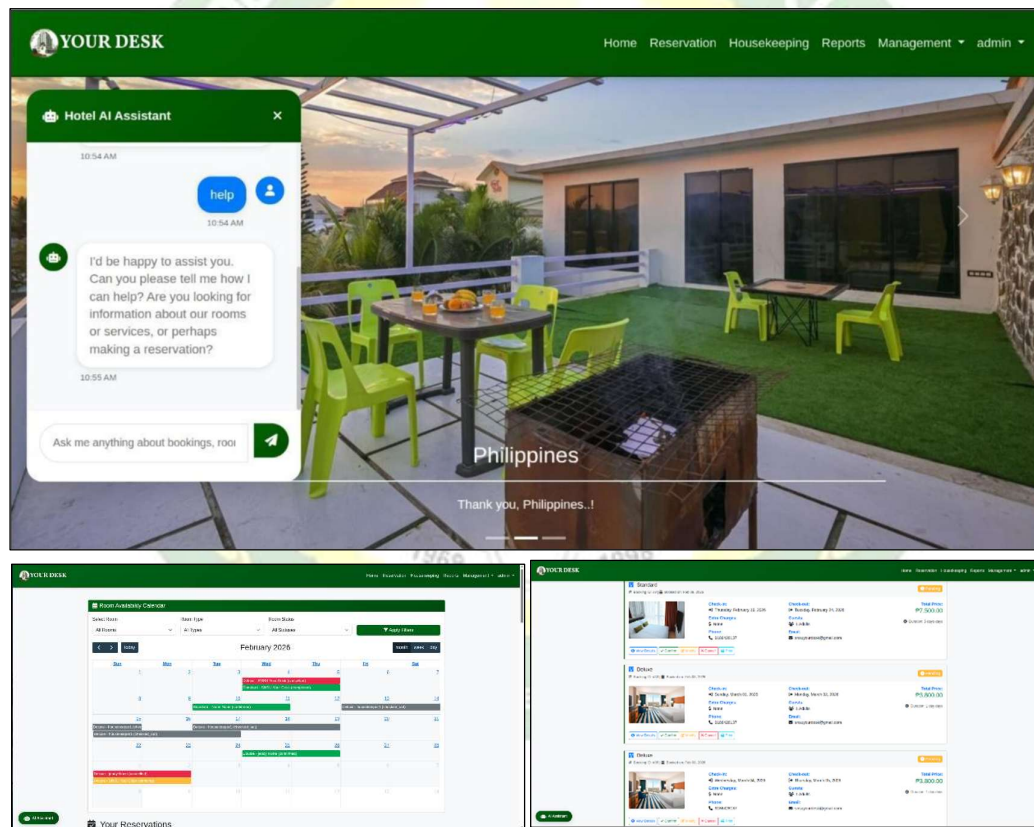


Figure 6. Reservation with Hotel AI Assistant Interface

This features in Figure 6 fulfills the second objectives and its corresponding scope, which focuses on the implementation of calendar-based reservation synchronization with a local AI hotel assistant. The system centralizes reservation schedules into a single integrated calendar that updates in real time. This allows staff to

monitor bookings without switching between multiple records or platforms. The local AI hotel assistant supports daily operations by responding to queries and assisting in managing reservations. It also provides automated reminders to guests and staff which helps reduce missed bookings and late arrivals. Another key function is reservation conflict detection which identifies overlapping schedules and prevents double booking. This improves accuracy in reservation handling and enhances overall efficiency in hotel operations. Moreover, the web-based access shown in Figure 6 allows various users to navigate the system functionalities without installing additional applications, making the system both accessible and secure. The seamless recordkeeping and efficient reservation process demonstrates the system's commitment to enhancing guest experience, better decision making while also maintaining ease of use for the stakeholders.

5.3. Inventory Tracking and Maintenance Scheduling

The system monitors resource usage and availability while enabling organized scheduling and tracking of maintenance and housekeeping activities, as shown in the Inventory, Maintenance and Housekeeping Interface on Figure 7.

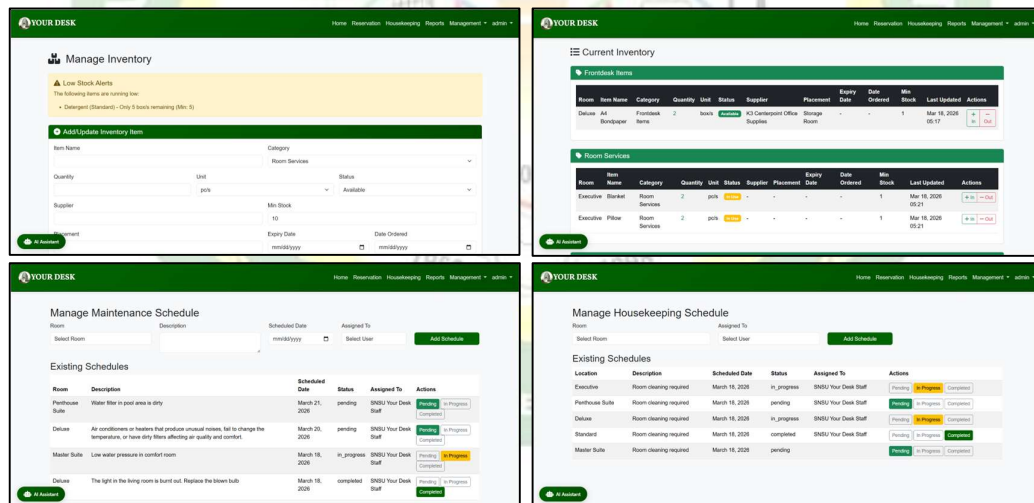


Figure 7. Inventory, Maintenance, and Housekeeping Interface

This components in Figure 7 directly addresses the study's third objective, which focuses on monitoring inventory and managing maintenance scheduling within the HM laboratory and front desk office. The system tracks inventory usage and updates availability in real time. This helps staff know which items are ready for use and which need replenishment. It reduces the risk of shortages and improves resource planning.

The feature also allows scheduling of maintenance activities for rooms and laboratory facilities. Staff can record tasks and monitor their completion without confusion. This ensures that facilities remain functional and safe for daily operations and prevents rooms from being occupied despite the unavailability due to maintenance. In addition, the housekeeping interface allows housekeepers to easily view assigned tasks and update the status of make-up rooms and cleaning processes. This provides real-time monitoring that allows the system to automatically mark cleaned rooms as available for reservation, generating more profit and seamless operations. By streamlining inventory control and maintenance coordination, the system enhances overall efficiency and supports better service delivery.

5.4. Automated Reports and Data Analytics

The system enables efficient data retrieval for operational report generation and provide basic analytics with downloadable CSV files to support monitoring and decision making, as shown in the Reports Interface on Figure 8.

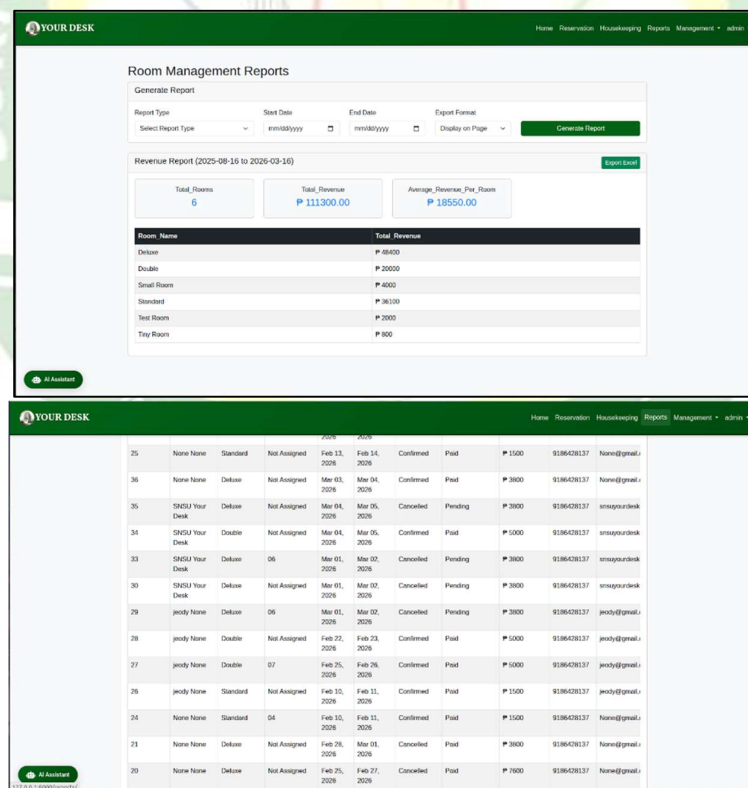


Figure 8. Reports Interface

This features in Figure 8 fulfills the fourth objective by providing a comprehensive system for generating automated operational reports, including reservation logs, inventory reports, and maintenance records, which significantly reduces manual reporting workload. The system organizes data in a structured format that enables staff to quickly access relevant information without the need to search through multiple sources or files. It also provides basic data analytics that allow management to identify trends, monitor performance, and make informed decisions based on accurate, real-time information from different hotel operations. The downloadable CSV report function supports further data analysis outside the system, making it easier for staff to integrate information into other tools or prepare presentations. By automating report generation, the system minimizes human error and ensures consistency across all operational records, improving overall reliability and transparency in monitoring activities. Users can customize report parameters to focus on specific timeframes, report type, and export file format, enhancing flexibility and relevance in operational evaluation. The interface also allows instant access to historical records, which supports trend analysis and aids in strategic planning for future operations. Overall, the feature strengthens decision-making processes by providing clear, actionable insights while saving time and improving efficiency in report management. By combining reporting and analytics, the system effectively enhances operational oversight, supports management objectives, and promotes data-driven performance improvements.

Chapter 6

CONCLUSION AND RECOMMENDATIONS

6.1. Conclusion

The development of “Your Desk” system provided a secure and efficient approach to managing front office operations and enhancing HM front desk training simulation of Surigao del Norte State University. The system achieved all four specific objectives by integrating reservation management, inventory and maintenance tracking, AI-assisted operational support, and automated reporting with analytics to streamline daily operations and training processes. These features collectively replace the limitations of manual reservation processes and eliminate the need for trial versions of front-desk platforms, allowing students to engage with actual hotel industry system technologies. It ensured that guests, staff, and housekeepers experienced smooth coordination, timely notifications, and accurate updates regarding reservations, room availability, and operational tasks. The system demonstrated reliability through its real-time data updates, automated conflict detection, and user-friendly interface, which minimized errors and enhanced workflow efficiency. Additionally, the integration of a local AI assistant provided support in operational decision-making and improved task management without requiring constant supervision. The automated reports and downloadable analytics further strengthened monitoring and evaluation, allowing administrators and students to assess performance effectively. Overall, “Your Desk” system successfully bridged the gap between theoretical learning and practical application, providing a comprehensive, interactive, and modern platform for hospitality management education. The implementation of this system highlights the potential of technology to improve operational accuracy, resource management, and student preparedness for real-world hotel environments.

6.2. Recommendations

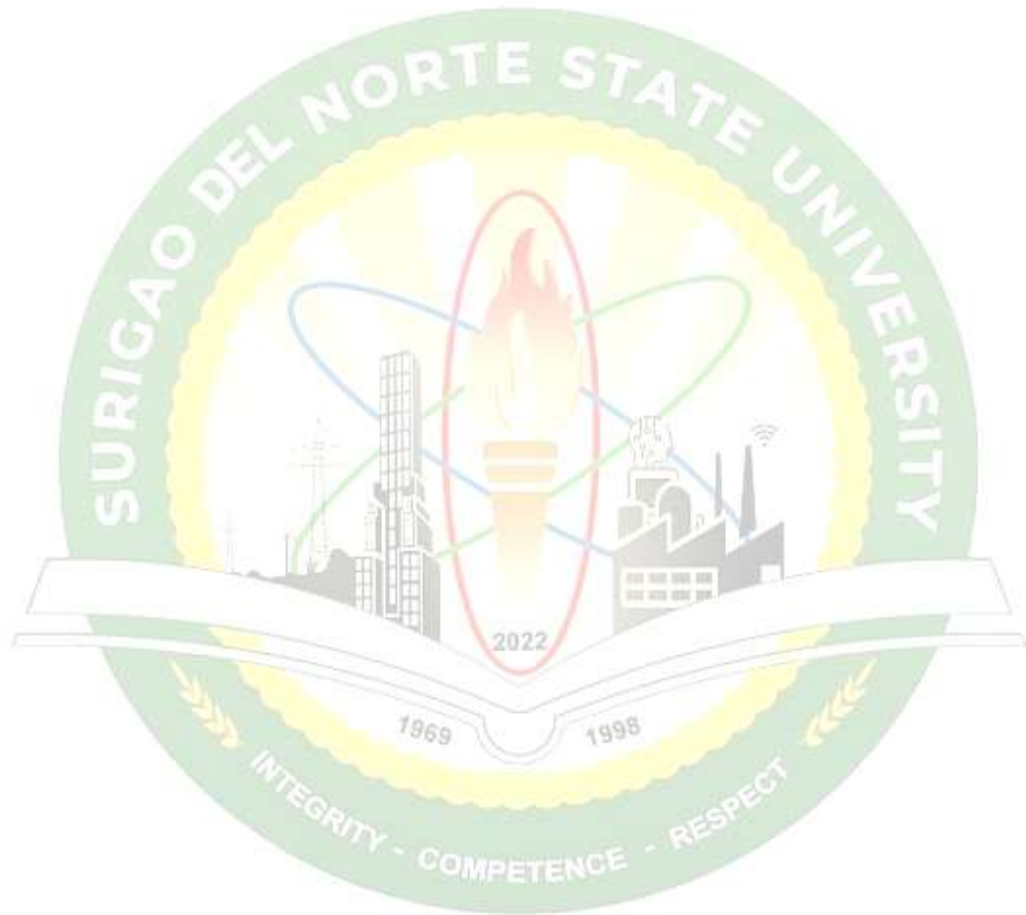
The researchers suggest several enhancements to increase the scalability and utility of “Your Desk” Front Office Operations and Management System for the academic community and hotel establishments. The study focused primarily on the design and technical development, as well as the platform’s initial deployment in the university setting and on free cloud hosting web applications. Because the development followed Agile Software Development methodology in a short period of

implementation time, the majority of the timeline was dedicated to creating the functional prototype. This allowed testing essential core functionalities, including reservation management, inventory tracking, maintenance scheduling, and automated reporting features. Consequently, a formal evaluation of user acceptance was not conducted during this cycle to prioritize the technical stability of critical system features. Although the researchers administered a system usability scale questionnaire to stakeholders to gain feedback for system improvement and accuracy, the number of respondents was insufficient based on the total number of HM and CS students and instructors in the university to justify comprehensive usability conclusions. It is recommended that future studies continue this work by conducting a formal evaluation stage to collect quantitative data on the system's impact on administrative efficiency and student learning outcomes. Expanding the application's reach beyond the initial pilot groups will help ensure it is fully optimized for adoption across various academic and hotel institutions. Based on the findings and technical limitations encountered during the development phase, the following specific recommendations are provided for future researchers to ensure the long-term sustainability of the front office operations management solution. These suggested improvements focus on technical expansion, enhanced user experience, cloud optimization, and addressing current environmental and hardware constraints identified during testing, ensuring the system can scale effectively and remain relevant for broader implementation.

1. Conduct a comprehensive evaluation process using system usability scale questionnaires to provide a statistical baseline for user satisfaction and overall system effectiveness.
2. Expand the data analytics in reports page to include predictive insights, additional export file formats, and pre-designed templates for easier decision-making.
3. Enhance the maintenance module by applying algorithms that automate maintenance scheduling, optimize task prioritization, and prevent conflicts in room or equipment availability.
4. Add a feature for real-time messaging across user accounts and user group chats to improve communication between guests, staff, housekeepers, and

administrators, ensuring faster coordination, shorter query response time and task updates.

5. Implement a phased rollout across all colleges and departments to test the system's scalability and its ability to handle large volumes of simultaneous user data.



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APPENDICES

Appendix A

LETTER TO CONDUCT THE STUDY



Appendix B

LETTER ASKING PERMISSION TO FLOAT THE QUESTIONNAIRE/ CONDUCT INTERVIEW



Appendix C

DOCUMENTATION DURING DATA GATHERING



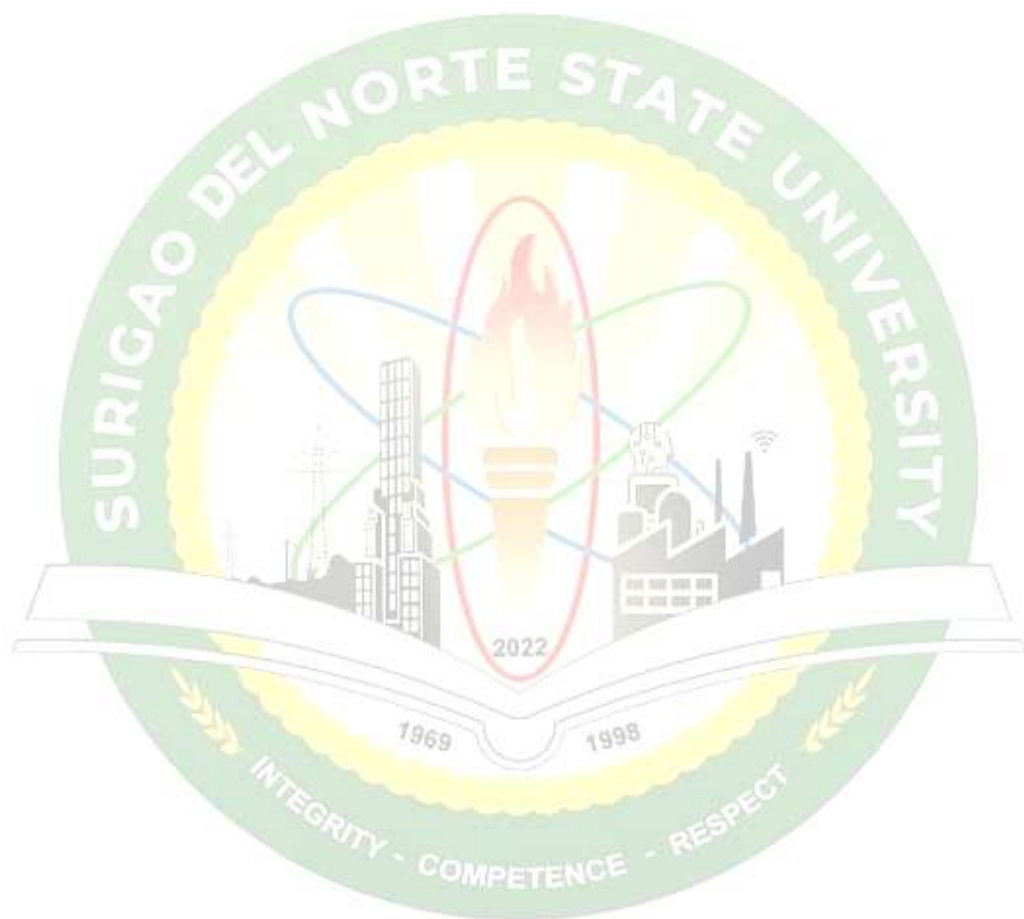
Appendix D

LETTER ASKING PERMISSION TO DEPLOY THE SYSTEM



Appendix E

MINUTES OF THE PROPOSAL DEFENSE AND FINAL DEFENSE



Appendix F
CERTIFICATE OF DEPLOYMENT



Appendix G
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